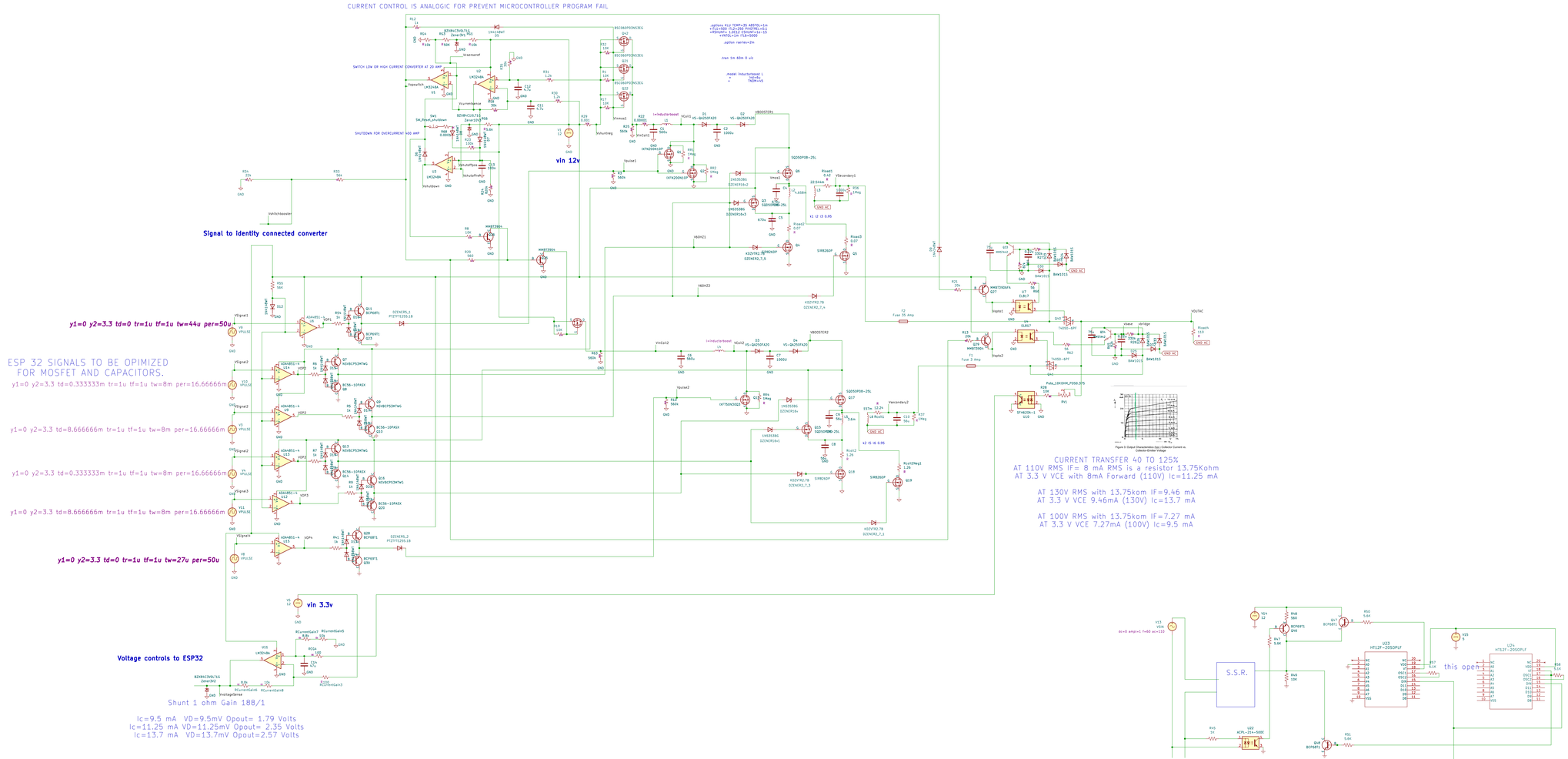
CURRENT CONTROL IS ANALOGIC FOR PREVENT MICROCONTROLLER PROGRAM FAIL
SWITCH LOW OR HIGH CURRENT CONVERTER AT 20 AMP
SHUTDOWN FOR OVERCURRENT 40 AMP
vin 12v
Signal to identity connected converter
y1=0 y2=3.3 td=0 tr=1u tf=1u tw=44u per=50u
ESP 32 SIGNALS TO BE OPIMIZED
FOR MOSFET AND CAPACITORS.
y1=0 y2=3.3 td=0.333333m tr=1u tf=1u tw=8m per=16.66666m
y1=0 y2=3.3 td=8.666666m tr=1u tf=1u tw=8m per=16.66666m
y1=0 y2=3.3 td=0.333333m tr=1u tf=1u tw=8m per=16.66666m
y1=0 y2=3.3 td=8.666666m tr=1u tf=1u tw=8m per=16.66666m
y1=0 y2=3.3 td=0 tr=1u tf=1u tw=27u per=50u
vin 3.3v
Voltage controls to ESP32
Shunt 1 ohm Gain 188/1
Ic=9.5 mA VD=9.5mV Opout= 1.79 Volts
Ic=11.25 mA VD=11.25mV Opout= 2.35 Volts
Ic=13.7 mA VD=13.7mV Opout=2.57 Volts
CURRENT TRANSFER 40 TO 125%
AT 110V RMS IF= 8 mA RMS is a resistor 13.75Kohm
AT 3.3 V VCE with 8mA Forward (110V) Ic=11.25 mA
AT 130V RMS with 13.75kom IF=9.46 mA
AT 3.3 V VCE 9.46mA (130V) Ic=13.7 mA
AT 100V RMS with 13.75kom IF=7.27 mA
AT 3.3 V VCE 7.27mA (100V) Ic=9.5 mA
S.S.R.
POWER SUPPLY
TO CONTROLLED
DEVICE
this open
RECEIVER RX-99V3 CIRCUT
A1,A2,A3,A4,A5,A6,A7 D9, D10, D11 ARE HARDWIRED
TO DEFINE THE ADDRESS OF THE DEVICE.
A0=1 OF TURN ON , A0=0 TURN OFF
FOR BATTERY CHARGE WITH 3 DEVICES:
CONDITION ABS(VIN - VOUT) < 1 V (BUCK-BOOST CONVERTER)
NEXT TWO CONDITIONS ARE BY SOFTWATER FORCING OFF MOSFETS
VOUT
CONTROL REMOTE CIRCUIT FOR THE ALGORITHM DEVICE NUMBER = ON , DEVICE NUMBER +1 = OFF
WHICH USE ODD AN EVEN TO ACT OVER THE PULSE FOR ON OR OFF AND 8 BITS FOR CODING.
TO ESP32 IC2
S.S.R.
POWER SUPPLY
TO CONTROLLED
DEVICE
RECEIVER RX-99V3 CIRCUT
A1,A2,A3,A4,A5,A6,A7 ARE HARDWIRED
TO DEFINE THE ADDRESS OF THE DEVICE.
A0=1 OF TURN ON , A0=0 TURN OFF
D9, D10, D11
VOLTAGE OF MCP SIGNAL MUST BE SAME AS TE OR LESS
AD8 MUST BE CONNECTED WITH MCP23017 PORTB0.
AD9, AD10, AND AD11 MUST BE DISCONNECTED OR USED WITH A LED
DEINDING OF USAGE OF OPTION 1 OR 2 OF DECODER
TRANSCEIVER TX-99V3 CIRCUT
Added circuit for control remote.
from previous design simplified by adding more
signals in microcontroller.
Darle Martin Lobos (patent pending)
Sheet: /
File: NEW BOOSTER4.kicad_sch
Title: Battery Charger and DC/AC converter
Size: A2 Date: 2025-03-20 Rev: 2
KiCad E.D.A. 9.0.7-unknown Id: 1/1